

No. of Functional Features Included in the Solution

Date	04 July 2026
Team ID	
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	5 Marks

Functional Features Overview

S. No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Weather Data Input Form	Collects 10 weather/rainfall parameters from the user	templates/index.html	Done	High
2	Flood Prediction Engine	Runs the trained Random Forest model on submitted inputs	predict.py	Done	High
3	Instant Result Display	Shows "No Flood" or "Flood Likely" immediately after submission	app.py, index.html	Done	High
4	Model Training Pipeline	Trains and evaluates the Random Forest model from the dataset	train_model.py	Done	Medium
5	Model Persistence	Saves/loads the trained model using Joblib for reuse	train_model.py, predict.py	Done	Medium
6	Data Preprocessing	Reads and prepares the Excel rainfall dataset for training	train_model.py	Done	Medium

Feature Summary

Metric	Count / Value
Total Features Planned	6
Total Features Implemented	6
Core / Must-Have Features	5
Additional / Nice-to-Have Features	1
Features Tested & Verified	6

Feature Category Breakdown

S. No	Category	Features in Category	Example Features
1	User Interface (UI)	1	Weather Data Input Form
2	Backend / Logic	2	Flood Prediction Engine, Instant Result Display

S. No	Category	Features in Category	Example Features
3	Database / Storage	1	Data Preprocessing (Excel dataset)
4	API / Integration	0	None currently (planned: weather API)
5	Machine Learning	2	Model Training Pipeline, Model Persistence